Name: ______________________

Writing Equations from Real World Situations

1. What is the unknown? Define your variable.
2. How would you solve the problem in your head?
3. Use the inverse operation to write an equation.
4. Solve the equation showing all work.

Situation:	Define Variable:	Write equation and solve:
Alicia had money in her savings account. She made a withdrawal of \$60 and her balance was then \$198. Write an equation that can be used to find the starting amount of money in her savings account.	m= starting \$ **How would you solve?** Add **Inverse operation?** subtract	$\boxed{m - \cancel{60}} = \boxed{198}$ $\cancel{+60}$ +60 m = \$258
A bag of erasers is split between 3 students. Each student receives 6 erasers. How many erasers were in the bag?	e = erasers in bag **How would you solve?** multiply **Inverse operation?** divide	$\frac{3}{1} \cdot \boxed{\frac{e}{3}} = \boxed{6 \cdot 3}$ e = 18
Five friends went to dinner on Friday night. If the total bill was \$60 and the friends split the bill evenly, write an equation to find the amount each person would pay.	p= \$ ea. person pays **How would you solve?** divide **Inverse operation?** multiply	$\boxed{\cancel{5}p} = \boxed{60}$ $\frac{\quad}{\cancel{5}\,1}$ $\frac{\quad}{5}$ p = 12

Situation: The temperature dropped 11° in the last hour. The temperature is currently 50° F. Write an equation that could be used to find what the starting temperature was one hour ago. **Define Variable:** t= starting temp **What's happening to variable?** dropping	**Write equation and solve:** $\cancel{t-11} = 50$ $\cancel{+11}$ $+11$ $\boxed{t = 61°}$	**Situation**: Inara had some quarters in her purse to buy stickers. She and her 3 friends evenly shared her quarters. Each of the girls received 2 quarters. Write an equation to find how many quarters were in the purse to begin with. **Define Variable:** q = quarters in purse **What's happening to variable?** being divided	**Write equation and solve:** $\frac{\cancel{4}}{1} \cdot \frac{q}{\cancel{4}} = 2 \cdot 4$ $\boxed{q = 8}$
Situation: On Saturday, Taco walked 1.25 miles through the neighborhood. By the end of the weekend, he had walked a total of 3.6 miles. Write an equation to find the distance Taco walked on Sunday. **Define Variable:** d= Sunday's dist. **What's happening to variable?** Combined w/ Sat.	**Write equation and solve:** $d \cancel{+1.25} = 3.6$ $\cancel{-1.25}$ -1.25 $\boxed{d = 2.35}$ $3.\overset{5}{\cancel{6}}\cancel{0} - 1.25 = 2.35$	**Situation**: Yesterday, Freya drank 12 ounces of lemonade in the afternoon. By the end of the day she drank a total of 28 fluid ounces of lemonade. Write an equation that would represent, m, the ounces of lemonade she drank in the morning. **Define Variable:** m = morning oz. **What's happening to variable?** added to afternoon	**Write equation and solve:** $m \cancel{+12} = 28$ $\cancel{-12}$ -12 $\boxed{m = 16}$
Situation: Mrs. Grimm signed up to bring two dozen (24) cupcakes to the school bake sale. She has a pan that makes 6 cupcakes at once. How many pans will she need to make all the cupcakes at once? **Define Variable:** P= pans needed **What's happening to variable?** Multiply pans	**Write equation and solve:** $\frac{\cancel{6}p}{\cancel{6}} = \frac{24}{6}$ $\boxed{p = 4}$	**Situation**: Alexis had $180 in her savings account. She made a deposit and her balance was then $213. Write an equation that can be used to determine the amount she deposited in her account. **Define Variable:** d= $ deposited **What's happening to variable?** Added to 180	**Write equation and solve:** $\cancel{180} + d = 213$ $\cancel{-180}$ -180 $\boxed{d = 33}$